UNIVERSITY OF TWENTE.

VALIDATING BIOMASS PRODUCTS IN SUMATRA'S TRANSFORMATION LANDSCAPE

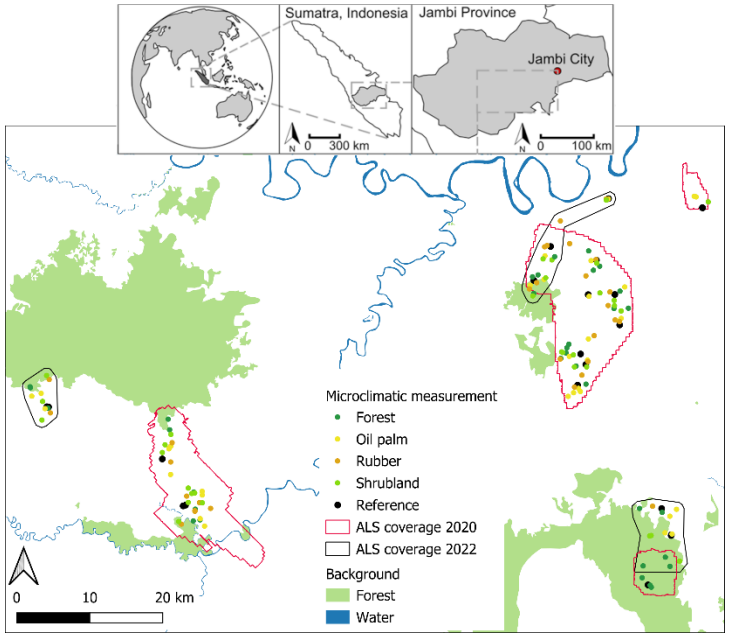
MICHAEL SCHLUND

ITC FACULTY OF GEO-INFORMATION SCIENCE AND EARTH OBSERVATION

e-mail m.schlund@utwente.nl

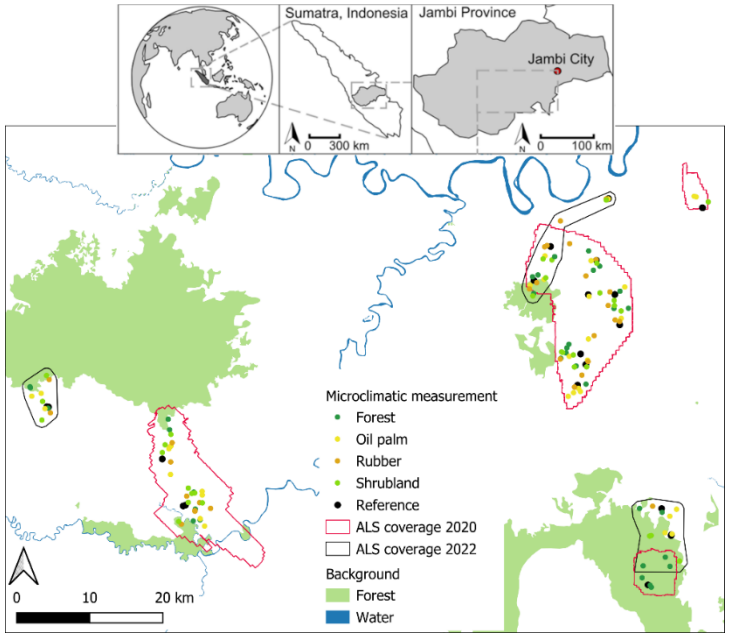
STUDY AREA & DATA

- Lowlands of Jambi, Sumatra, Indonesia
  - Extensive studies since 2012 through Collaborative Research Center (Ecological and Socioeconomic Functions of Tropical Lowland Rainforest Transformation Systems)
  - Dynamic mosaic of land-use systems



STUDY AREA & DATA

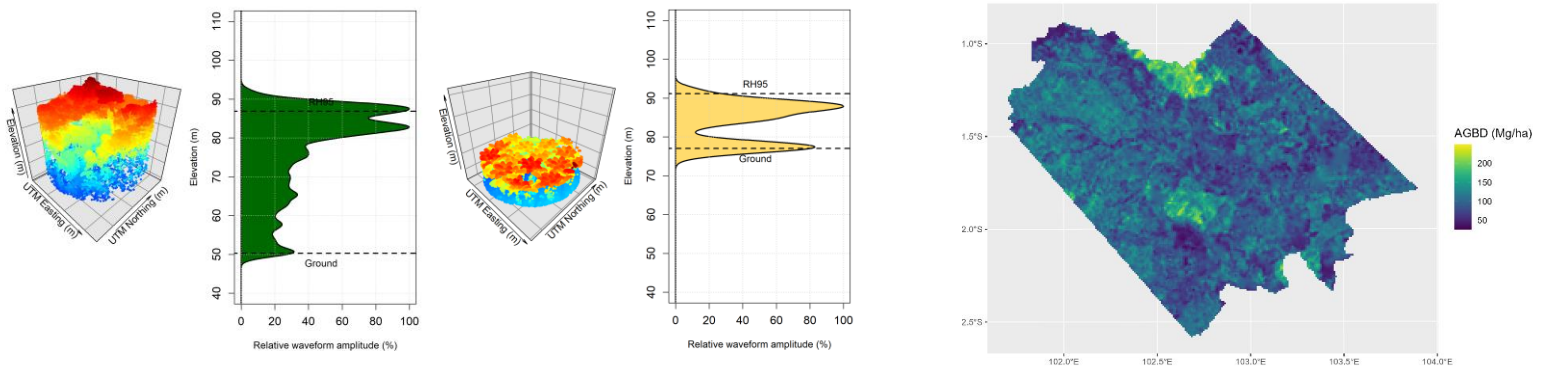
- Lowlands of Jambi, Sumatra, Indonesia
  - >100 field plots
  - 2 Airborne Laser Scanning campaigns
  - Terrestrial and mobile laser scanning
  - Remote sensing data to support understanding impact of transformation



PREVIOUS STUDIES IN THE AREA

- We developed a hierarchical model relating spatial distribution of GEDI to field-estimated AGBD
- Hierarchical model extendable with spatially complete auxiliary data (e.g. BIOMASS)

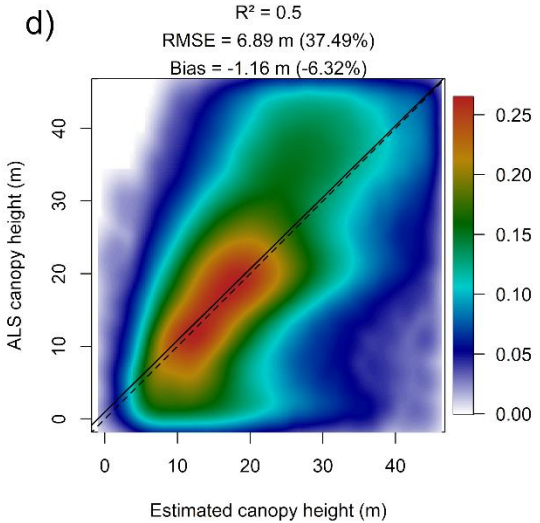
Plot AGBD → Simulated RH95 → Observed RH95 → AGBD predictions

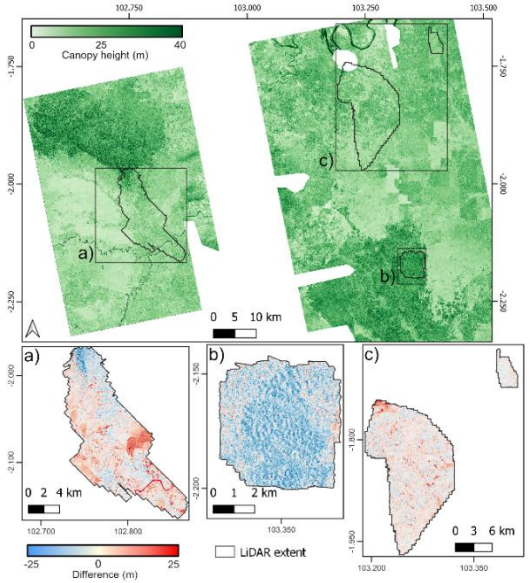


May et al. (2024) https://doi.org/10.1016/j.rse.2024.114384

PREVIOUS STUDIES IN THE AREA

- TanDEM-X achieved moderate accuracy in forest height estimation
- Acquisition conditions and vegetation structure with impact on accuracy

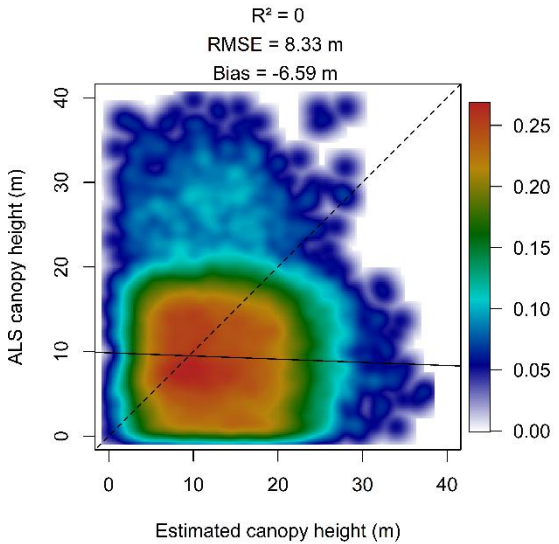


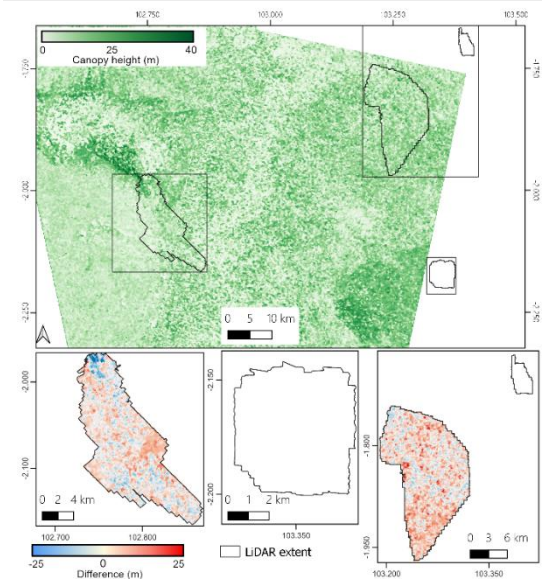


Schlund et al. (2023) https://doi.org/10.1111/2041-210X.13933

PRELIMINARY BIOMASS FOREST HEIGHT RESULTS

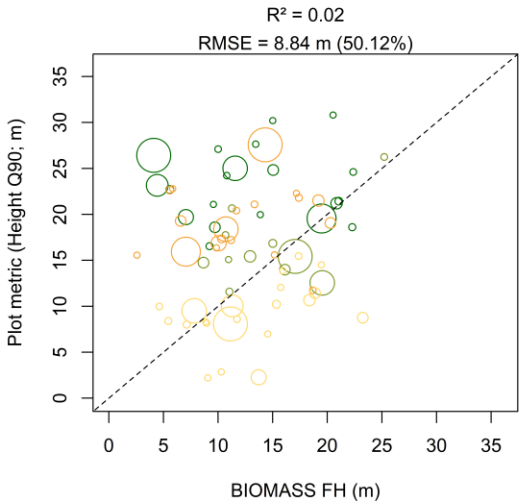
- BIOMASS Forest Height data available since October 2026
- Data for large intact forest area not available

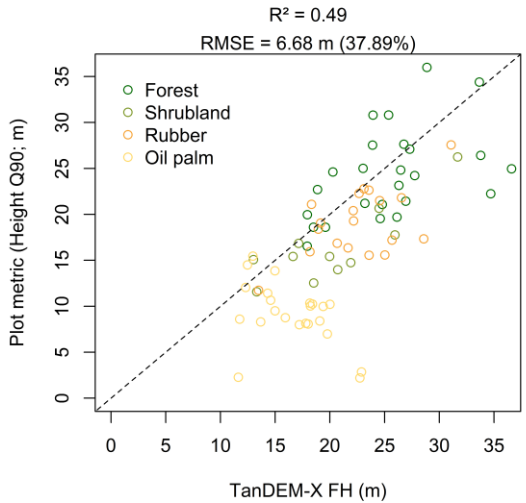




PRELIMINARY BIOMASS FOREST HEIGHT RESULTS

- BIOMASS Forest Height data available since October 2026
- Relatively low performance requires further investigation





* Size corresponds to quality